

End Semester Examination – May 2025

Course: CSE2025 – Python Programming
Programme: B.Tech
Semester: II

Duration: 3 hrs.

Max. Marks: 40

Instructions:

- All questions are compulsory.
- Missing data, if any, can be suitably assumed.
- Use of non-scientific calculator is allowed.

S.No	Question	Marks
1	Write a program to accept five integers from the user, one at a time, and display them in ascending order.	3
2	Write a program to count the number of even numbers in the tuple $t = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$.	3
3	Write a program to accept a line of integers separated by spaces from the user and display all unique integers.	4
4	Write a program to check if a given string is a valid email address. A valid email should contain both '@' and '.' characters.	4
5	Write a program to determine if a student has passed or failed based on marks in three subjects. The student passes if the average of the three subjects is at least 40 and has at least 33 in each subject. Assume each subject is out of 100.	5
6	Write a program to assign grades to a student based on their mark in a subject according to the following scheme: 90–100: A, 80–89: B, 70–79: C, 60–69: D, below 60: F. Accept the mark from the user.	4
7	Write a program to print the first n prime numbers, where n is input by the user.	4
8	Write a program to calculate the sum of all even numbers between 1 and n , where n is input by the user.	4
9	Write a function to reverse a string. Then, use this function in a program to check if a string input by the user is a palindrome.	4
10	Write a program to read a text file named 'input.txt', count the frequency of each word, and write the results to another file named 'output.txt'. Assume that words are separated by spaces.	6